Re-evaluating Multimodal LLMs for Influenza Forecasting

A 28-Model Benchmark of Accuracy, Peak Skill, Parameter Efficiency and Statistical Evidence

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Final Year Project • Progress Presentation

Four evaluation axes ask different questions

#	Dimension	Evidence	Coverage
1	Accuracy	MSE and MAE	28 models
2	Peak skill	Timing, height and joint screen	28 models
3	Parameter efficiency	MSE versus parameter count	11 models
4	Statistical evidence	Paired tests with proper Holm	17 models

Historical results, with explicitly different analysis subsets

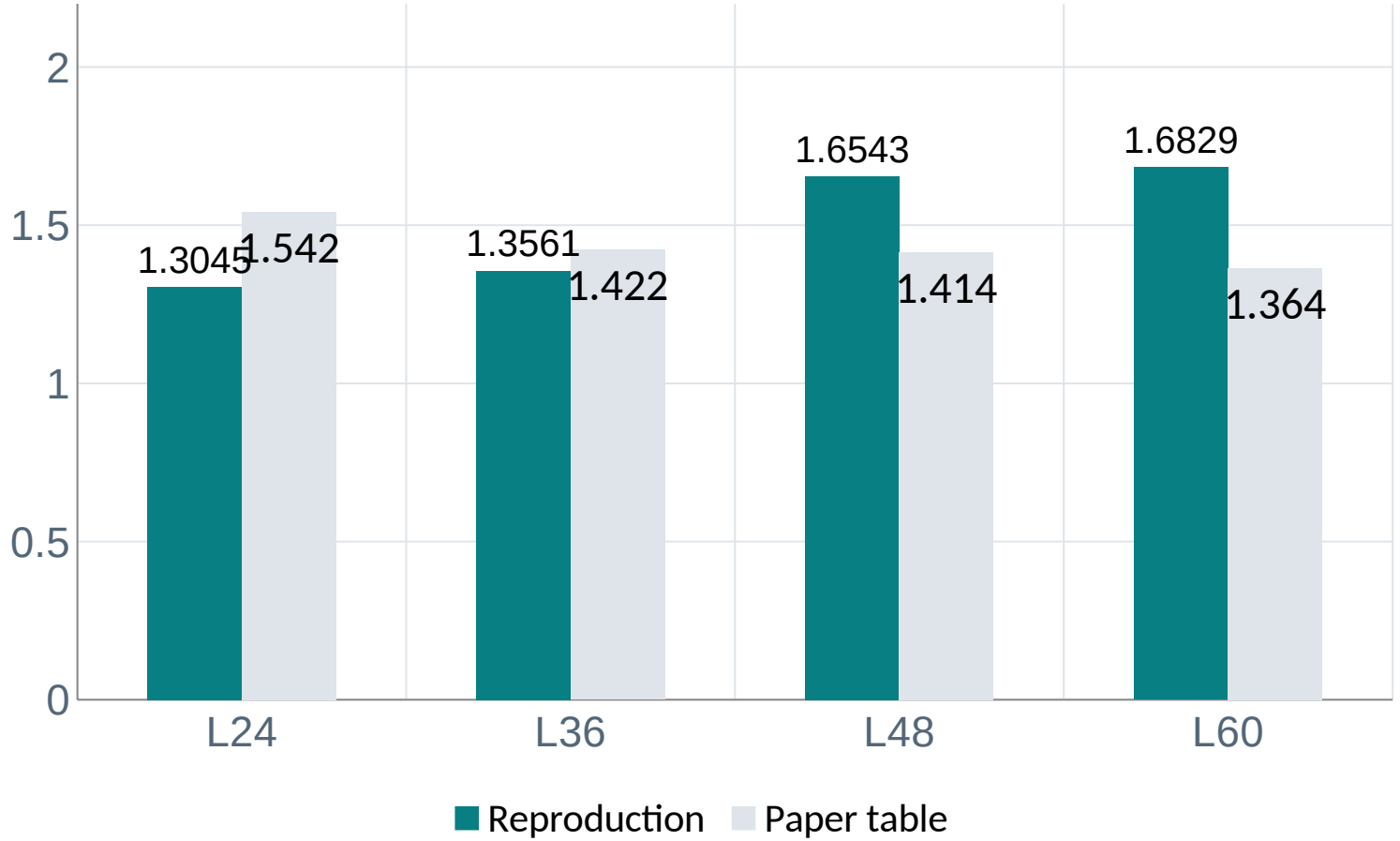
Reproduction and benchmarking use separate run sets

Purpose	Reproduction	Benchmark
Seeds	2021–2030 (10)	42 / 43 / 44 (3)
MIFlu mean MSE	1.4995	1.5574
Scored windows	160 / 144 / 144 / 128	Same windows across arms

28 retained arms = 13 conventional + 4 LLM-based + 11 frozen encoders

966 rows, 7 channels. Split 676 / 97 / 193. New common scoring remains unresolved.

Reproduction differs by horizon and error measure



MSE mean

1.4995 / 1.4355

+4.5%

Two lower, two higher

MAE mean

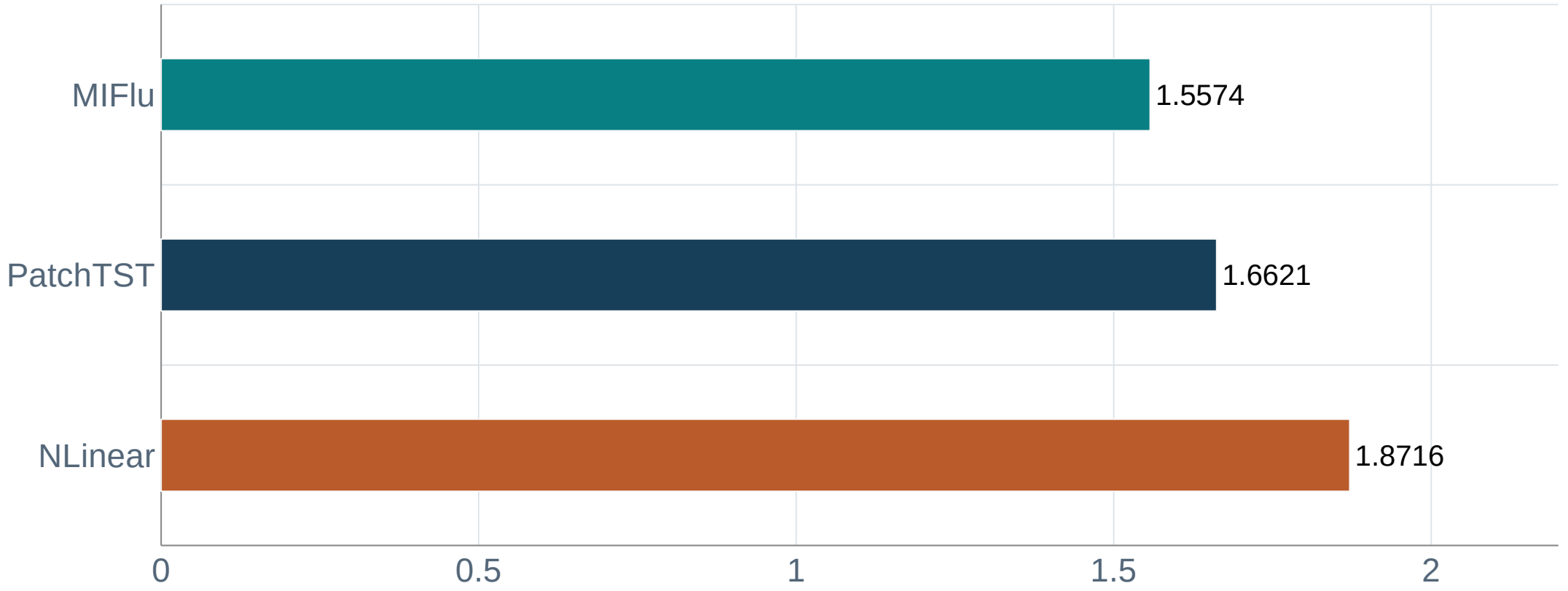
0.8043 / 0.7452

+7.9%

One lower, three higher

Source: reproduction CSV. Raw ten-seed runs are not independently rescored here.

MIFlu leads the observed average-error ranking



Mean MAE: MIFlu 0.8199 PatchTST 0.8786 NLinear 0.9183

Lower is better. Leading three shown; the full 28-model table is in the internal report.

Peak skill separates timing from height

Joint screen: PTE ≤ 1 week AND PIE ≤ 25%

Model	Mean PTE (weeks)	Mean PIE (%)	Joint passes
MIFlu	1.128	26.846	4/16
PatchTST	1.240	32.755	0/16
NLinear	1.677	9.477	4/16

MIFlu has lower mean timing error. NLinear has lower mean height error.

Project joint mean-error screen. Each denominator is 4 horizons × 4 leads.

Equal pass counts hide different passing cells

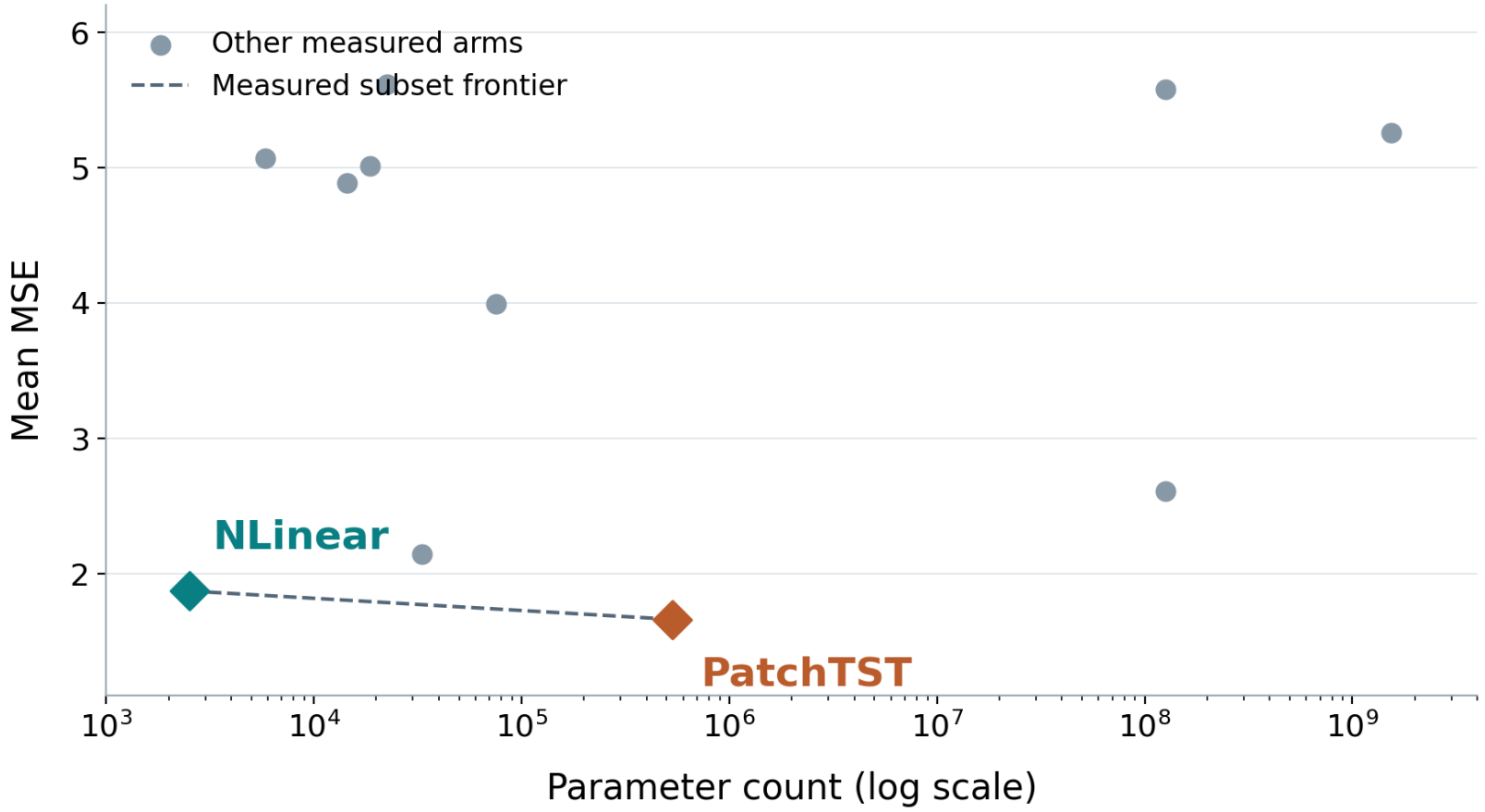


MSE order: MIFlu < PatchTST < NLinear

PIE order: NLinear < MIFlu < PatchTST

Each green cell passes both thresholds. s0–s3 are forecast weeks 1–4.

The parameter frontier exposes a size–error trade-off



211×

PatchTST / NLinear parameter count

11.2%

lower observed MSE

11 exported arms

$R^2 = 0.0223$

Subset frontier. MIFlu and encoders excluded. Parameter count is a size proxy, not measured runtime.

Correct Holm supports no familywise rejection

Before correction

128 / 136

Pairs with $p < 0.05$ in the declared family

After proper Holm

0 / 136

NLinear–PatchTST $p \approx 0.1172$

Observed gain remains. Equivalence is unproven.

17 models including MIFlu. DLinear and encoders excluded.

Minimum adjusted $p \approx 0.0664$; exact-test resolution and dependence limit inference.

Frozen-feature encoders underperform in this protocol

Frozen encoder + trained linear readout

MSE 4.7374–5.2049

All 11 exceed decision tree MSE 4.3198

Each encoder passes 0/16 joint cells

Scope: these feature/readout configurations. Generation, prompting, fine-tuning and numerical foundation models remain untested.

CPU tests confirm the inspected no-text invariant

Default author_faithful path

Time positions

Later text

Causal attention prevents time positions from reading later text

Forecast head

Head reads time positions only

Evidence status

Fixed-weight CPU tests support text invariance

Three assertions passed

Zero skipped

Ablation queue 60/60 DONE

Per-variant analysis pending

Supplied CPU execution record. Training gradients and independent training trajectories remain unverified.

The ten-seed expansion remains in progress

1. Core outputs complete: 160/160
2. National matrix in progress: 141/1120*
3. Harmonise scoring, then validate regions

Optional O2 remains validation-only recalibration

*141/1120: owner report, 5 Nov 00:15 HKT; package 117/1120. O2: 0 new GPU training if suitable validation/test exports exist; CPU cost unmeasured.

Model choice depends on the evaluation objective

MIFlu: lowest observed average MSE

NLinear: lower peak-height error at low parameter count

Corrected tests: no familywise rejection at this resolution

Next step: harmonise scoring, then validate across origins and regions